

MPM1D – Exam Review – Units 1-2 Homework

1. Simplify each of the following by writing a single base and/or getting rid of all brackets, do not calculate.

a. $9^4 \times 9^5$ $\underline{9^9}$
 $4+5$

c. $3^9 \div 3^3$ $\underline{3^6}$
 $9-3$

e. $(2^8)^2$ $\underline{2^{16}}$
 8×2

g. $7^6 \div 7^2 \times 7^5$ $\underline{7^9}$
 $6-2+5$
 $= 4+5$
 $= 9$

b. $5^8 \times 5^4 \times 5$ $\underline{5^{13}}$
 $8+4+1$

d. $7^8 \div 7^4 \div 7^0$ $\underline{7^4}$
 $8-4-0$

f. $(8^2)(8^4) \div 8^5$ $\underline{8^1 \rightarrow 8}$
 $2+4-5$

h. $4^{12} \div (4^3)^3 \cdot 4^0$ $\underline{4^3}$
 $12-9+0$
 $= 3+0$
 $= 3$

2. Simplify each of the following by writing as a single base, do not calculate.

a. $\left(\frac{2}{3}\right)^8 \left(\frac{2}{3}\right)^2 \div \left(\frac{2}{3}\right)^4$
 $= \left(\frac{2}{3}\right)^{8+2-4}$
 $= \left(\frac{2}{3}\right)^6$

b. $\frac{3^{16} \cdot (3^3)^3 \cdot 3}{3^{11} \times (3^2)^3}$
 $= \frac{3^{16} \cdot 3^9 \cdot 3}{3^{11} \cdot 3^6}$
 $= \frac{3^{16+9+1}}{3^{11+6}}$
 $= \frac{3^{26}}{3^{17}}$
 $= 3^9$

3. State the degree of each of the following polynomials.

a. $12x^2y^4z^{14}$ $\underline{\text{Degree is } 20}$
 $2+4+14$

b. $13ab + 12cd - 2a^2b^3$ $\underline{\text{Degree is } 5}$
 $2 \quad 2 \quad 5$

c. $8 + y - 3m + 17y^4 - 8m^2n + 4ab$ $\underline{\text{Degree is } 4}$
 $0 \quad 1 \quad 1 \quad 4 \quad 3 \quad 2$

4. Identify each of the following polynomials as a monomial or binomial or trinomial.

a. $\underline{3a^2b} - \underline{7b^2c} - \underline{18}$ 3 terms

Trinomial

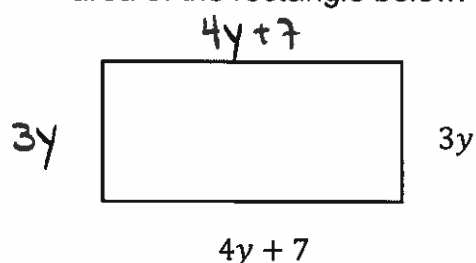
b. $\underline{4c} - \underline{8a}$ 2 terms

Binomial

c. $\underline{24a^3b^4c^2d}$ 1 term

Monomial

5. Write a simplified expression for the perimeter and a simplified expression for the area of the rectangle below. Show your work. Recall $P = \sum \text{sides}$



$$A = LW$$

$$P = \sum \text{sides}$$

$$= 3y + 4y + 7 + 3y + 4y + 7$$

$$= 14y + 14 \text{ units}$$

$$A = LW$$

$$= (3y)(4y + 7)$$

$$= 12y^2 + 21y \text{ units}^2$$

6. Simplify each expression below. Show your work.

a. $\underline{6x} + 9y + 8y - \underline{5x}$

$$= 6x - 5x + 9y + 8y$$

$$= x + 17y$$

b. $\underline{12a^2b} - \underline{9ab^3} + \underline{17a^2b} - \underline{7ab^3} + 9ab^2$

$$= 12a^2b + 17a^2b + 9ab^2 - 9ab^3 - 7ab^3$$

$$= 29a^2b + 9ab^2 - 16ab^3$$

c. $27 - \underline{5a} - \underline{14b^2} + 41 + \underline{7a} - \underline{19b^2} + \underline{13a}$

$$= -14b^2 - 19b^2 - 5a + 7a + 13a + 27 + 41$$

$$= -33b^2 + 15a + 68$$

d. $5(y + 4)$

$$= 5y + 20$$

e. $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $1(4y - 7) - 1(y - 8) - 1(-7y + 3)$

$$= 4y - 7 - y + 8 + 7y - 3$$

$$= 10y - 2$$

f. $\downarrow \quad \downarrow \quad \downarrow$
 $2y(7y^2 - 3y + 4)$

$$= (2y)(7y^2) + (2y)(-3y) + (2y)(4)$$

$$= 14y^3 - 6y^2 + 8y$$

g. $5y^3(2y^2 - 6ay)$

$$= (5y^3)(2y^2) - (5y^3)(-6ay)$$

$$= 10y^5 + 30ay^4$$

$$\begin{aligned}
 \text{h. } & -2y^2(-5y^5 + 7y - 4a + y^4) \\
 & = (-2y^2)(-5y^5) + (-2y^2)(7y) + (-2y^2)(-4a) + (-2y^2)(y^4) \\
 & = 10y^7 - 14y^3 + 8ay^2 - 2y^6
 \end{aligned}$$

$$\begin{aligned}
 \text{i. } & 2y(4 - 8y) - 3(2y^2 + 6) \\
 & = 8y - 16y^2 - 6y^2 - 18 \\
 & = 8y - 22y^2 - 18 \\
 & = -22y^2 + 8y - 18
 \end{aligned}$$

$$\begin{aligned}
 \text{j. } & \frac{-(3m^2n^5)(6m^5n^4)}{(2m^2n)^3} \\
 & = \frac{-[(3)(6)(m^2)(m^5)(n^5)(n^4)]}{(2)^3 (m^2)^3 (n)^3} \\
 & = \frac{-18m^7n^9}{8m^6n^3} \\
 & = -\frac{9mn^6}{4}
 \end{aligned}$$

Unit 2 – Equations

Checks

1. Solve each equation below and check your answer.

$$\begin{aligned}\frac{x}{7} &= 21 \\ \frac{147}{7} &= 21 \\ 21 &= 21 \\ \text{LS} &= \text{RS} \\ \text{☺}\end{aligned}$$

$$\begin{aligned}9 + \frac{x}{5} &= 3 \\ 9 + \frac{-30}{5} &= 3 \\ 9 + (-6) &= 3 \\ 9 - 6 &= 3 \\ 3 &= 3 \\ \text{LS} &= \text{RS} \\ \text{☺}\end{aligned}$$

$$\begin{aligned}9 &= \frac{x-8}{2} \\ 9 &= \frac{26-8}{2} \\ 9 &= \frac{18}{2} \\ 9 &= 9 \\ \text{LS} &= \text{RS} \\ \text{☺}\end{aligned}$$

$$\begin{aligned}\frac{x}{4} &= \frac{x-5}{2} \\ \frac{10}{4} &= \frac{10-5}{2} \\ \frac{5}{2} &= \frac{5}{2} \\ \text{LS} &= \text{RS} \\ \text{☺}\end{aligned}$$

$$\begin{aligned}\text{a. } \frac{x}{7} &= 21 \\ 7\left(\frac{x}{7}\right) &= 7(21) \\ \frac{7x}{7} &= 7(21) \\ x &= 147\end{aligned}$$

$$\begin{aligned}\text{c. } 9 + \frac{x}{5} &= 3 \quad -9 \\ \frac{x}{5} &= -6 \\ 5\left(\frac{x}{5}\right) &= 5(-6) \\ x &= -30\end{aligned}$$

$$\begin{aligned}\text{e. } 9 &= \frac{x-8}{2} \quad (2) \\ 18 &= x-8 \quad +8 \\ 26 &= x \\ x &= 26\end{aligned}$$

$$\begin{aligned}\text{g. } \frac{x}{4} &= \frac{x-5}{2} \\ 4\left(\frac{x}{4}\right) &= 4\left(\frac{x-5}{2}\right) \\ x &= \frac{4x-20}{2} \\ x &= 2x-10 \quad -2x \\ -x &= -10 \quad -1 \\ x &= 10\end{aligned}$$

$$\begin{aligned}\text{b. } 4x - 15 &= 13 \\ +15 \quad +15 \\ 4x &= 28 \\ \frac{4x}{4} &= \frac{28}{4} \\ x &= 7\end{aligned}$$

$$\begin{aligned}\text{d. } 4(x-6) &= -52 \\ 4x-24 &= -52 \\ +24 \quad +24 \\ 4x &= -28 \\ \frac{4x}{4} &= \frac{-28}{4} \\ x &= -7\end{aligned}$$

$$\begin{aligned}\text{f. } 3x - 8 &= -2x + 32 \\ +2x \quad +2x \\ 5x - 8 &= 32 \\ +8 \quad +8 \\ 5x &= 40 \\ \frac{5x}{5} &= \frac{40}{5} \\ x &= 8\end{aligned}$$

$$\begin{aligned}\text{h. } 7 &= \frac{3x+2}{5} \\ (5)(7) &= \left(\frac{3x+2}{5}\right)(5) \\ 35 &= \frac{15x+10}{5} \\ 35 &= 3x+2 \quad -2 \\ 33 &= 3x \\ \frac{33}{3} &= \frac{3x}{3} \\ x &= 11\end{aligned}$$

Checks

$$\begin{aligned}4x-15 &= 13 \\ 4(7)-15 &= 13 \\ 28-15 &= 13 \\ 13 &= 13 \\ \text{LS} &= \text{RS} \\ \text{☺}\end{aligned}$$

$$\begin{aligned}4(x-6) &= -52 \\ 4(-7-6) &= -52 \\ 4(-13) &= -52 \\ -52 &= -52 \\ \text{LS} &= \text{RS} \\ \text{☺}\end{aligned}$$

$$\begin{aligned}3x-8 &= -2x+32 \\ 3(8)-8 &= -2(8)+32 \\ 24-8 &= -16+32 \\ 16 &= 16 \\ \text{LS} &= \text{RS} \\ \text{☺}\end{aligned}$$

$$\begin{aligned}7 &= \frac{3x+2}{5} \\ 7 &= \frac{3(11)+2}{5} \\ 7 &= \frac{33+2}{5} \\ 7 &= \frac{35}{5} \\ 7 &= 7 \\ \text{LS} &= \text{RS} \\ \text{☺}\end{aligned}$$

2. Solve for the unknown variable and check your answer.

$$\frac{3}{4}(x+2) = \frac{2}{3}(x-5)$$

$$\frac{3(x+2)}{4} = \frac{2(x-5)}{3}$$

$$\frac{3x+6}{4} = \frac{2x-10}{3}$$

$$12 \left(\frac{3x+6}{4} \right) = 12 \left(\frac{2x-10}{3} \right)$$

$$\frac{36x+72}{4} = \frac{24x-120}{3}$$

$$9x+18 = 8x-40$$

$$x+18 = -40$$

$$x = -58$$

$$\frac{3}{4}(x+2) = \frac{2}{3}(x-5)$$

$$\frac{3}{4}(-58+2) = \frac{2}{3}(-58-5)$$

$$\frac{3}{4} \cdot \frac{-56}{1} = \frac{2}{3} \cdot \frac{-63}{1}$$

$$-\frac{168}{4} = -\frac{126}{3}$$

$$-42 = -42$$

$$LS = RS$$

☺

3. Write an expression for each one of the following statements. Let x be the unknown number.

a. Nineteen more than a number

$$\underline{x+19}$$

b. One-quarter of a number decreased by three

$$\underline{\frac{x}{4}-3 \text{ or } \frac{1}{4}x-3}$$

c. Seven less than twice a number

$$\underline{2x-7}$$

d. The sum of five a number is divided by four

$$\underline{\frac{x+5}{4}}$$

4. Isolate for A in each of the following equations. Show your work.

a. $\frac{AB}{B} = \frac{C}{B}$

$$A = \frac{C}{B}$$

b. $\frac{AD}{D} + \frac{B}{-B} = \frac{C}{-B}$

$$\frac{AD}{D} = \frac{C-B}{D}$$

$$A = \frac{C-B}{D}$$

c. $\frac{A^2}{B} = C$

$$B\left(\frac{A^2}{B}\right) = B(C)$$

$$A^2 = BC$$

$$\sqrt{A^2} = \sqrt{BC}$$

$$A = \sqrt{BC}$$

d. $\frac{A}{C} - \frac{B}{+B} = \frac{D}{+B}$

$$\frac{A}{C} = \frac{B+D}{B}$$

$$C\left(\frac{A}{C}\right) = C\left(\frac{B+D}{B}\right)$$

$$A = C\left(\frac{B+D}{B}\right)$$

$$A = BC + DC$$

e. $\frac{\sqrt{A}}{+B} - \frac{B}{+B} = \frac{C}{+B}$

$$\sqrt{A} = B+C$$

$$(\sqrt{A})^2 = (B+C)^2$$

$$A = (B+C)^2$$

f. $\frac{BC}{A} = D$

$$A\left(\frac{BC}{A}\right) = A(D)$$

$$\frac{BC}{D} = \frac{AD}{D}$$

$$A = \frac{BC}{D}$$

5. Write an expression for each description below. Don't forget to include let statements to describe your variables.

a. On a Math Contest, students receive 7 marks for each correct answer and lose 3 marks for each incorrect answer.

$$T = 7x - 3y$$

let T be total # of points
 " x " # of correct answers
 " y " " " wrong "

b. A team plays 50 games in a season. They can either win or lose, but there are no ties. The team earns 3 points for every win and loses 1 point for every loss.

$$T = 3W - L$$

let T be total points
 let W be # of games won
 " L " " " " lost

6. Mr. T works in a clothing store in the summer. He earns \$130 per day and a bonus of \$7 for every shirt that he sells and a bonus of \$9 for every pair of pants that he sells. Let E be his total earnings (in \$), let x be the number of shirts he sells, and let y be the number of pairs of pants he sells.

- a. Write an equation to show Mr. T's earnings in one day.

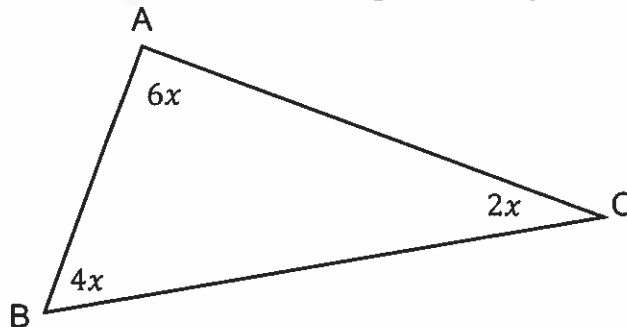
$$E = 130 + 7x + 9y$$

- b. Use your equation to determine how much Mr. T would earn if he sold 5 pairs of pants and 6 shirts in one day. Show your work. 5

$$\begin{aligned} E &= 130 + 7x + 9y \\ &= 130 + 7(6) + 9(5) \\ &= 130 + 42 + 45 \\ &= 217 \end{aligned}$$

$\therefore$ He would earn \$217.

7. Solve for the measure of each angle. Show your work.



$$6x + 4x + 2x = 180 \quad (\text{ASTT } \smile)$$

$$\frac{12x}{12} = \frac{180}{12}$$

$$x = 15^\circ$$

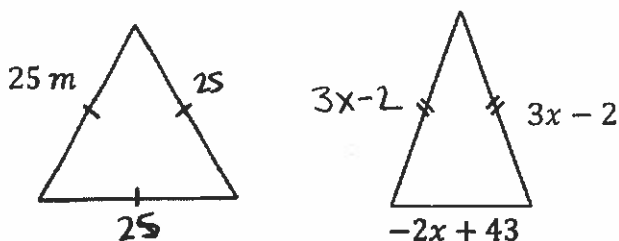
$$\begin{aligned} 2x &= 2(15) \\ &= 30^\circ \end{aligned}$$

$$\begin{aligned} 4x &= 4(15) \\ &= 60^\circ \end{aligned}$$

$$\begin{aligned} 6x &= 6(15) \\ &= 90^\circ \end{aligned}$$

$\therefore$ The angles are 30° , 60° , & 90°

8. The following two triangles have the same perimeter in metres. Determine the dimensions of the three side lengths of the second triangle.



$$P_1 = P_2$$

$$25 + 25 + 25 = \underline{3x-2} + \underline{3x-2} + \underline{-2x+43}$$

$$75 = 4x + 39$$

$$\frac{4x}{4} = \frac{36}{4}$$

$$x = 9 \text{ m}$$

$$\begin{array}{rcl} 3x-2 & & -2x+43 \\ = 3(9)-2 & & = -2(9)+43 \\ = 27-2 & & = -18+43 \\ = 25\text{m} & & = 25\text{m} \end{array}$$

For each one of the following word problems, please complete the following:

- Write let statements with one variable
- Write an equation
- Solve the equation
- Check your solution
- Write a therefore statement to answer the question

9. River is five years younger than Mountain. If the sum of their ages is 51, how old is each individual?

Let x be Mountain's age. Let $x-5$ be River's age.

$$x + x - 5 = 51$$

$$x = 28$$

$$2x - 5 = 51$$

$$\begin{array}{l} x - 5 \\ = 28 - 5 \\ = 23 \end{array}$$

$$\frac{2x}{2} = \frac{56}{2}$$

$$x = 28$$

$\therefore$ River is 23 yrs old.
Mountain is 28 yrs old

10. April and June had a bouncing contest. If April bounced seven less times than twice the bounces of June, and their combined bounces totaled 383, how many times did each girl bounce?

Let x be the # of times June bounced
 " $2x-7$ " " " " " April "

$$x + 2x - 7 = 383$$

$$3x - 7 = 383$$

$$\frac{3x}{3} = \frac{390}{3}$$

$$x = 130$$

$$\begin{aligned} 2x - 7 &= 2(130) - 7 \\ &= 260 - 7 \\ &= 253 \end{aligned}$$

$\therefore$ June bounced 130 times
 April " 253 "

11. Three angles are supplementary. The second angle is twice the measure of the first angle. The third angle is 6° less than triple the first angle. Determine the measure of each of the three angles.

Let x be the first angle
 " $2x$ " Second "
 " $3x-6$ " Third "

$$x + 2x + 3x - 6 = 180 \quad \text{S.A.T.}$$

$$6x - 6 = 180$$

$$\frac{6x}{6} = \frac{186}{6}$$

$$x = 31^\circ$$

$\therefore$ The angles are 31° , 62° , and 87°

$$\begin{aligned} x &= 31^\circ & 2x &= 2(31) & 3x - 6 &= 3(31) - 6 \\ & & &= 62^\circ & &= 93 - 6 \\ & & & & &= 87^\circ \end{aligned}$$

12. The sum of four consecutive even integers is 268. Determine the value of the integers.

Let	x	be	the	first	#
"	$x+2$	"	"	second	"
"	$x+4$	"	"	third	"
"	$x+6$	"	"	fourth	"

$$\underline{x} + \underline{x+2} + \underline{x+4} + \underline{x+6} = 268$$

$$4x + 12 = 268$$

$$\frac{4x}{4} = \frac{256}{4}$$

$$x = 64$$

$x = 64$	$x+2$	$x+4$	$x+6$
	$= 64+2$	$= 64+4$	$= 64+6$
	$= 66$	$= 68$	$= 70$

$\therefore$ The #s are 64, 66, 68, and 70

Answer Key – Unit 1

1a) 9^9

1b) 5^{13}

1c) 3^6

1d) 7^4

1e) 2^{16}

1f) $8^1 \rightarrow 8$

1g) 7^9

1h) 4^3

2a) $\left(\frac{2}{3}\right)^6$

2b) 3^9

3a) 20

3b) 5

3c) 4

4a) trinomial

4b) binomial

4c) monomial

5) $P = 14y + 14 \text{ units}$

$A = 12y^2 + 21y \text{ units}^2$

6a) $x + 17y$

6b) $29a^2b + 9ab^2 - 16ab^3$

6c) $68 + 15a - 33b^2$

6d) $5y + 20$

6e) $10y - 2$

6f) $14y^3 - 6y^2 + 8y$

6g) $10y^5 - 30ay^4$

6h) $10y^7 - 14y^3 + 8ay^2 - 2y^6$

6i) $-22y^2 + 8y - 18$

6j) $-\frac{9mn^6}{4}$

Answer Key – Unit 2

1a) $x = 147$

1b) $x = 7$

1c) $x = -30$

1d) $x = -7$

1e) $x = 26$

1f) $x = 8$

1g) $x = 10$

1h) $x = 11$

2) $x = -58$

3a) $x + 19$

3b) $\frac{x}{4} - 3$

3c) $2x - 7$

3d) $\frac{5+x}{4}$

4a) $A = \frac{C}{B}$

4b) $A = \frac{C-B}{D}$

4c) $A = \sqrt{BC}$

4d) $A = BC + DC$

4e) $A = (B + C)^2$

4f) $A = \frac{BC}{D}$

5a) $T = 7x - 3y$ *ket*

5b) $T = 3x - y$ *ket*

6a) $E = 130 + 7x + 9y$

6b) \$217

7) $30^\circ, 60^\circ, 90^\circ$

9) 25 m

10) Mountain is 28 years old

River is 23 years old

11) June bounced 130 times

April bounced 253 times

12) $31^\circ, 62^\circ, 87^\circ$

13) 64, 66, 68, 70